

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry based Problem-Solving

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 – Industry based Problem-Solving	Assessment:	Assessment Tool 2– Industry based Problem-Solving
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
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Section / Batch:	2024- CSE(AI)	Date:	20-08-26

Code of Conduct

I __ S.Sirivally / 192472371 __ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __ Sirivally.S __

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. A cybersecurity system historically detects 95% of malicious activities. After deploying a new detection algorithm, it correctly detects 194 out of 200 attacks. Test whether the new algorithm has significantly improved the detection rate using a p-value.

Given

- Historical detection rate = **95%**
- New algorithm detects **194 out of 200**
- Sample proportion:

$$\hat{p} = \frac{194}{200} = 0.97$$

1. Population

All malicious attacks that the cybersecurity system could encounter.

2. Sample

200 attacks tested using the new algorithm.

3. Parameter

Population detection rate p .

4. Hypotheses

$$H_0 : p = 0.95$$

$$H_1 : p > 0.95$$

This is a right-tailed one-proportion test because we want to know whether detection has improved.

5. Significance level

$$\alpha=0.05$$

6. Test statistic

$$\begin{aligned} z &= \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \\ z &= \frac{0.97 - 0.95}{\sqrt{\frac{0.95(0.05)}{200}}} \\ z &\approx 1.30 \end{aligned}$$

7. p-value

$$\text{p-value} \approx 0.097$$

8. Decision

$$0.097 > 0.05$$

Therefore, do not reject H_0 .

9. Conclusion

There is not enough statistical evidence at the 5% significance level to conclude that the new cybersecurity algorithm has significantly improved the detection rate above 95%.

Industry interpretation

Although the observed detection rate increased from 95% to 97%, the improvement is not statistically significant based on this sample.

2. A hospital introduces a new appointment scheduling system. Before implementation, the average patient waiting time was 42 minutes. After implementation, a sample of 64 patients has an average waiting time of 38 minutes with standard deviation 12 minutes. Test whether the new system significantly reduces waiting time using the p-value approach.

Given

Historical mean waiting time:

$\mu_0=42$ minutes

Sample:

- $n=64$
- $\bar{x}=38$
- $s=12$

We want to determine whether waiting time decreased.

1. Population

All patients using the hospital appointment system.

2. Sample

64 patients after implementation.

3. Parameter

Population mean waiting time μ .

4. Hypotheses

$$H_0 : \mu = 42$$

$$H_1 : \mu < 42$$

This is a left-tailed one-sample t-test.

5. Significance level

$$\alpha=0.05$$

6. Test statistic

$$\begin{aligned}t &= \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \\t &= \frac{38 - 42}{12/\sqrt{64}} \\t &= \frac{-4}{1.5} \\t &= -2.667\end{aligned}$$

Degrees of freedom:

$$\text{Df} = 64 - 1 = 63$$

7. p-value

$$\text{p-value} \approx 0.0048$$

8. Decision

$$0.0048 < 0.05$$

Reject H_0 .

9. Conclusion

There is statistically significant evidence that the new appointment scheduling system reduces average patient waiting time.

Industry interpretation

The average waiting time decreased from 42 minutes to 38 minutes, and this reduction is statistically significant.

3. A social media platform claims that the average daily usage time of its users is 3 hours. A random sample of 100 users has an average usage time of 3.2 hours with standard deviation 0.8 hours. Determine whether the data provide sufficient evidence to reject the company's claim at $\alpha = 0.05$.

Given

Company claim:

$\mu=3$ hours

Sample:

- $n=100$
- $\bar{x}=3.2$
- $s=0.8$

The question asks whether the data provide evidence to reject the claim, so this is a two-tailed test.

1. Population

All users of the social media platform.

2. Sample

100 randomly selected users.

3. Parameter

Population mean daily usage time μ .

4. Hypotheses

$$H_0 : \mu = 3$$

$$H_1 : \mu \neq 3$$

5. Significance level

$$\alpha=0.05$$

6. Test statistic

$$t = \frac{3.2 - 3}{0.8/\sqrt{100}}$$
$$t = \frac{0.2}{0.08}$$
$$t = 2.50$$

Degrees of freedom:

$$df=99$$

7. p-value

For a two-tailed test:

$$p\text{-value} \approx 0.014$$

8. Decision

$$0.014 < 0.05$$

Reject H₀.

9. Conclusion

There is sufficient statistical evidence to reject the company's claim that average daily usage is 3 hours.

Industry interpretation

The sample suggests that actual average usage differs from 3 hours. Since the sample mean is 3.2 hours, the evidence indicates higher average usage.

4. A data science team develops a new classification model. The existing model has an accuracy of 82%, while the new model achieves 86% accuracy on a randomly selected test sample of 500 observations. Formulate suitable null and alternative hypotheses and determine whether the improvement is statistically significant using a p-value.

Given

Existing model accuracy:

$$p_0 = 0.82$$

New model:

- $n = 500$
- Accuracy = 86%

Number correctly classified:

$$500(0.86) = 430$$

Sample proportion:

$$\hat{p} = 0.86$$

We want to determine whether accuracy improved.

1. Population

All observations that could be classified by the new model.

2. Sample

500 randomly selected test observations.

3. Parameter

Population accuracy p of the new model.

4. Hypotheses

$$H_0: p = 0.82$$

$$H_1: p > 0.82$$

This is a right-tailed one-proportion test.

5. Significance level

$$\alpha = 0.05$$

6. Test statistic

$$z = \frac{0.86 - 0.82}{\sqrt{\frac{0.82(0.18)}{500}}}$$
$$z \approx 2.08$$

7. p-value

$$p\text{-value} \approx 0.019$$

8. Decision

$$0.019 < 0.05$$

Reject H_0 .

9. Conclusion

There is statistically significant evidence that the new classification model has higher accuracy than 82%.

Industry interpretation

The new model's observed accuracy increased from 82% to 86%, and the improvement is statistically significant at the 5% level.

5. A manufacturer claims that its production process has a defect rate of only 2%. From 2,000 randomly selected products, 55 are defective. Perform a hypothesis test at the 5% significance level using both (a) confidence interval and (b) p-value approaches. Compare the conclusions.

Given

Manufacturer's claimed defect rate:

$$p_0 = 0.02$$

Sample:

- $n = 2000$
- Defective products = 55

Sample defect rate:

$$\hat{p} = \frac{55}{2000} = 0.0275$$

So the observed defect rate is:

2.75%

The manufacturer's claim is 2%.

A. Hypothesis Test

Because the manufacturer says the defect rate is "only 2%", it is reasonable to test whether the actual rate is greater than 2%.

1. Population

All products manufactured by the company.

2. Sample

2,000 randomly selected products.

3. Parameter

Population defect rate p .

4. Hypotheses

$$H_0: p = 0.02$$

$$H_1: p > 0.02$$

5. Significance level

$$\alpha = 0.05$$

6. Test statistic

$$z = \frac{0.0275 - 0.02}{\sqrt{\frac{0.02(0.98)}{2000}}}$$
$$z \approx 2.40$$

7. p-value

For a right-tailed test:

$$p\text{-value} \approx 0.0083$$

8. Decision

$$0.0083 < 0.05$$

Therefore:

Reject H_0

9. Conclusion

There is statistically significant evidence that the actual defect rate is greater than 2%.

B. 95% Confidence Interval

For the sample proportion:

$$\hat{p} = 0.0275$$

Standard error:

$$SE = \sqrt{\hat{p}(1-\hat{p})}$$

$$SE \approx 0.00366$$

95% confidence interval:

$$\hat{p} \pm 1.96(SE)$$
$$0.0275 \pm 1.96(0.00366)$$

Approximately:

(0.0203,0.0347)

In percentage form:

2.03% to 3.47%

Confidence interval interpretation

The plausible range for the true defect rate is approximately 2.03%–3.47%.

Because 2% is below this interval, the confidence interval also provides evidence against the manufacturer's 2% claim.

Final industry conclusion

Both approaches lead to the same conclusion:

The evidence indicates that the production defect rate is significantly higher than the claimed 2%.
The manufacturer should investigate the production process and quality-control measures.